

Computer Networks

**Collection of autonomous computers
interconnected by a communication channel.**

- **Introduction**
- **Network Hardware**
- **Network Software**
- **Reference Models**
- **Example Networks**
- **Example Data Communication Services**
- **Network Standardization**

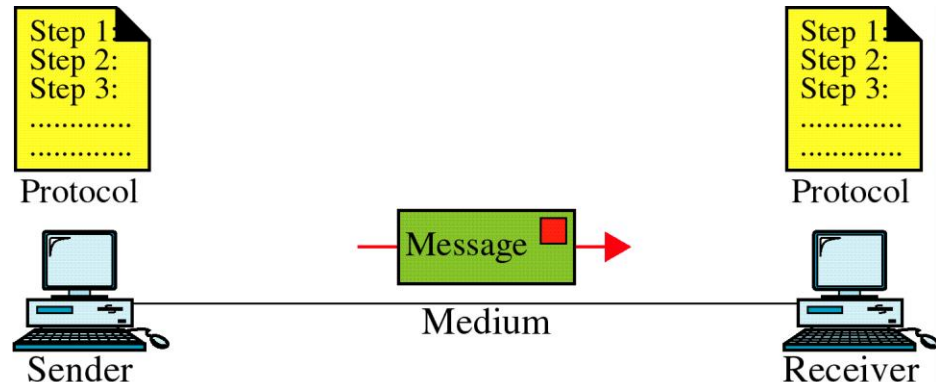
Introduction

Motivation for Networks

- **Information Access**
- **Sharing of Resources**
- **Facilitate Communications**

What A Network Includes

- **Transmission hardware**
- **Special-purpose hardware devices**
 - interconnect transmission media
 - control transmission
 - run protocol software
- **Protocol software**
 - encodes and formats data
 - detects and corrects problems



What A Network Does

- **Provides communication that is**
 - Reliable
 - Fair
 - Efficient
 - From one application to another
- **Automatically detects and corrects**
 - Data corruption
 - Data loss
 - Duplication
 - Out-of-order delivery
- **Automatically finds optimal path from source to destination**

Network Hardware

Transmission Technology:

Broadcast Link -

Single channel with multiple machines connected to it.

Point-to-point Link -

Individual connections between pairs of machines.

Packets -

Messages - the "chunk" of data transmitted from one machine to the next.

Addressing -

- + One to one: Packet contains specific target address.
- + Broadcasting: All machines on the network receive and process the packet.
- + Multicasting: A subset of machines receive and process the packet.

Types of Networks

LOCAL AREA NETWORKS

LANs are:

- Privately owned. Can be up to one kilometers long;
- Restricted so worst case transmission time can be contained.
- Run at speeds of 10, 100, or more Mbps.
- Low delay. High reliability.

Ethernet:

- IEEE 802.3.

- Bus based broadcast network with decentralized control at 10 or 100 Mbps.

Token Ring:

- IEEE 802.5

- Ring based broadcast network with token arbitration at 4 or 16 Mbps.

METROPOLITAN AREA NETWORKS:

MANS are:

- Larger version of LAN ("city" wide).

- Public or private / data or voice.

- Broadcast - no switches.

- Can be distinguished from LANs based on wiring mechanism.

WIDE AREA NETWORKS

WANS are: Networks spanning large distances.

Hosts or End Systems -

Machines running user applications.

(Communication) Subnet -

Connections between hosts - transmission lines + switches.

A "locality" understanding each other's addresses.

Circuits/Channels/Trunks -

Transmission lines move the bits.

Packet switching nodes/Intermediate systems/Data switching exchanges

Specialized computers moving data between several inputs
to several outputs.

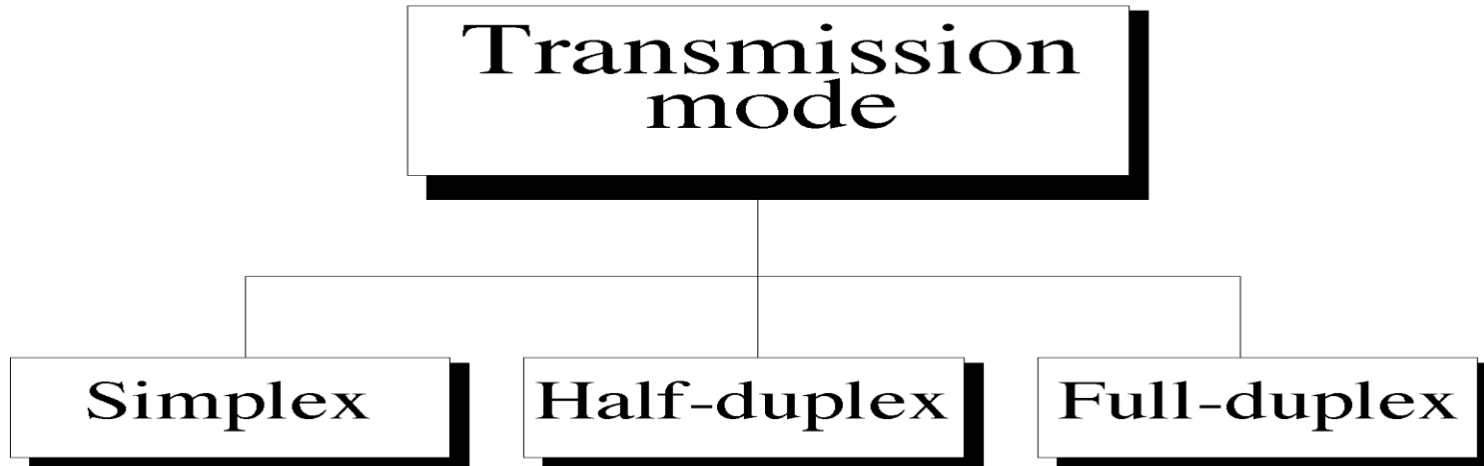
Router - Generic term for switching computers.

Point-to-point/Store-and-forward/Packet-switched -

Moving through a series of routers, packets are received at a router, stored there, then forwarded to the next router.

Transmission mode:

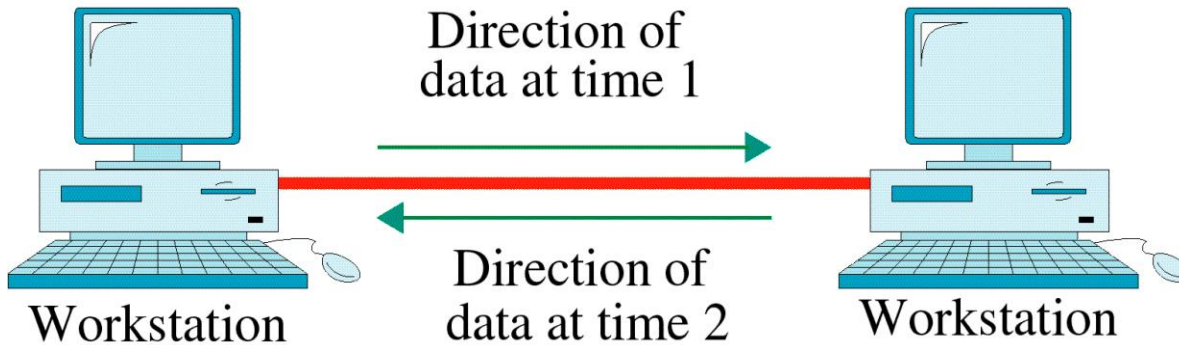
define the direction of signal flow between two linked devices.



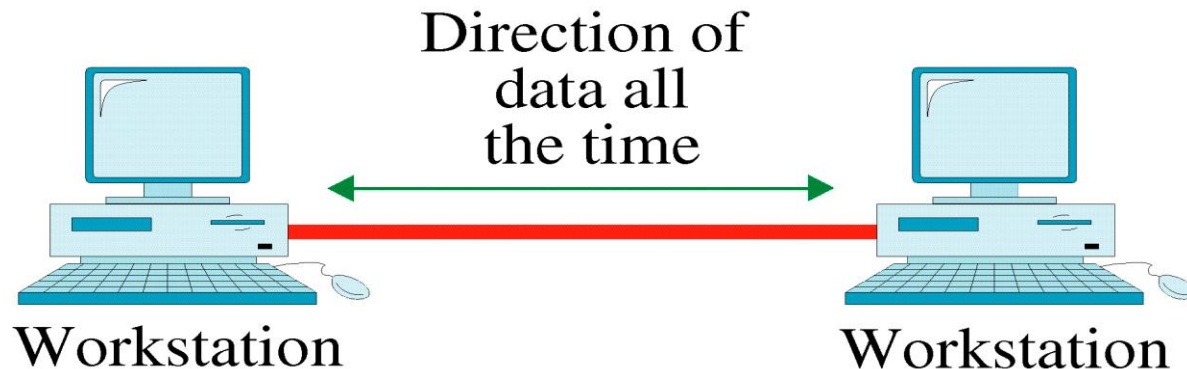
Simplex



Half-Duplex

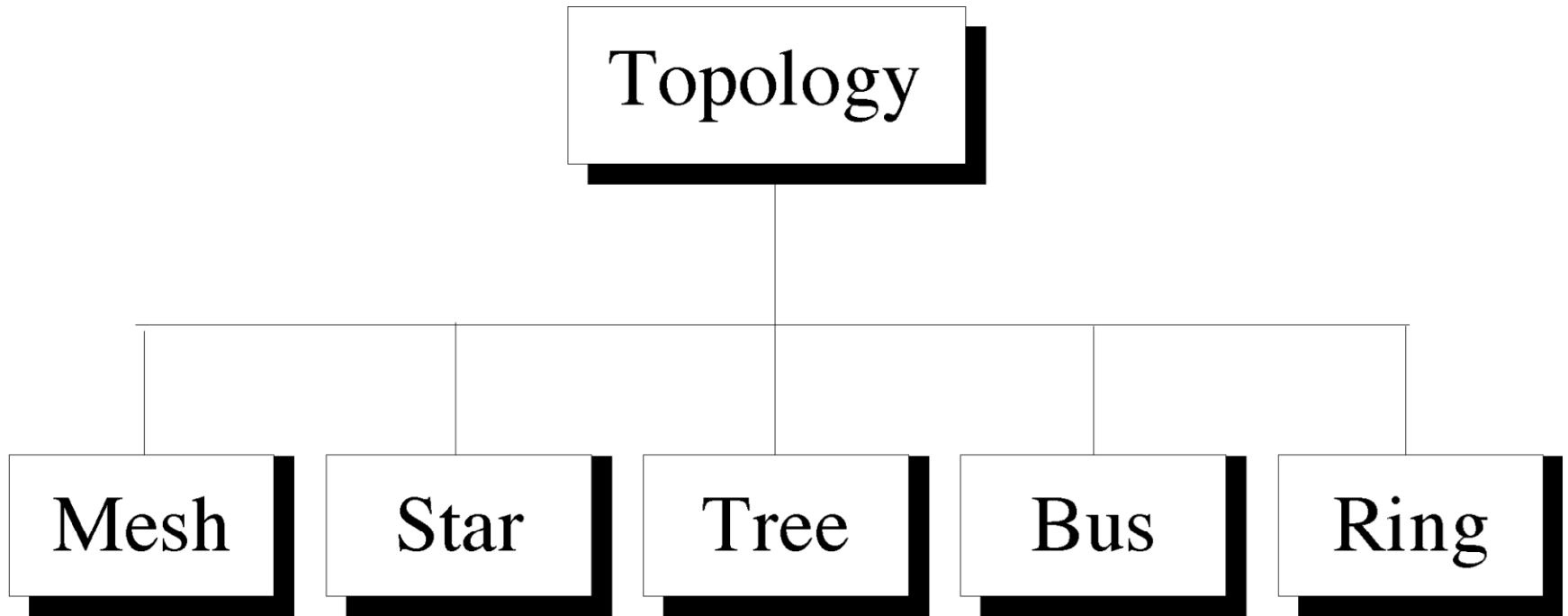


Full-Duplex



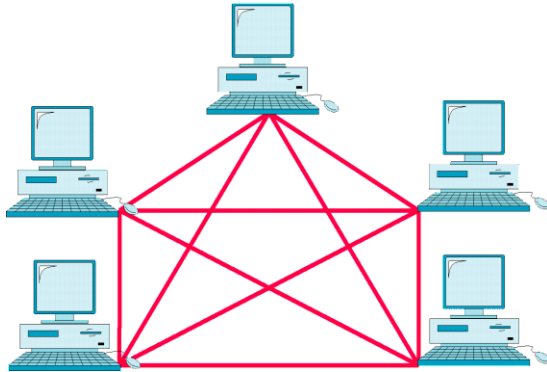
Topology

- Topology defines the physical or logical arrangement of links in a network

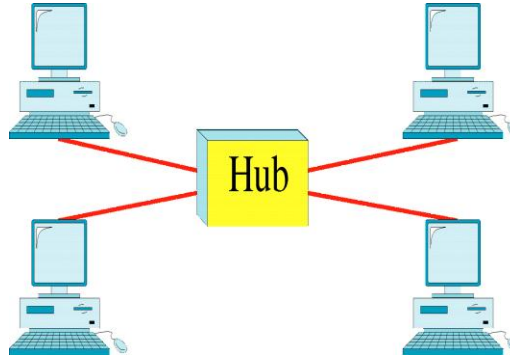


Topology

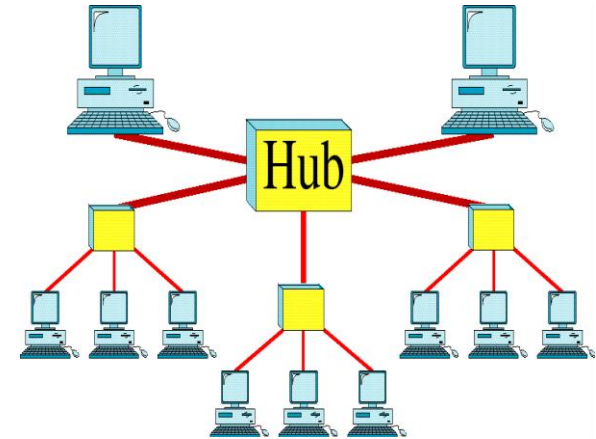
Mesh



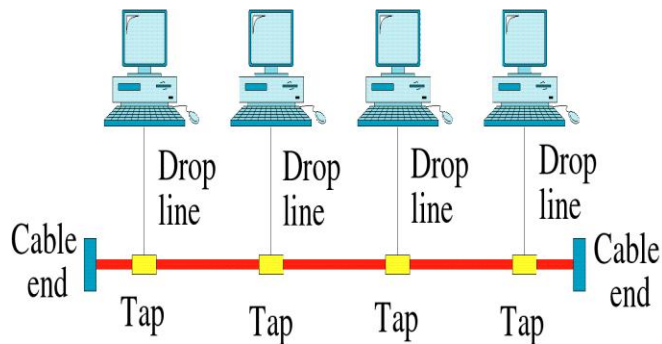
Star



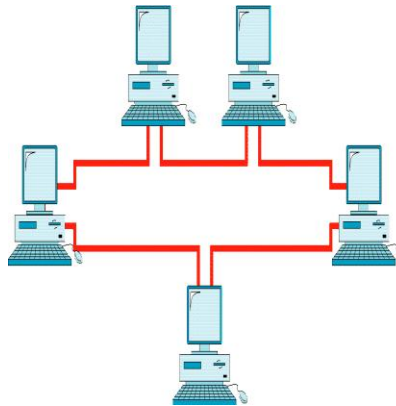
Tree



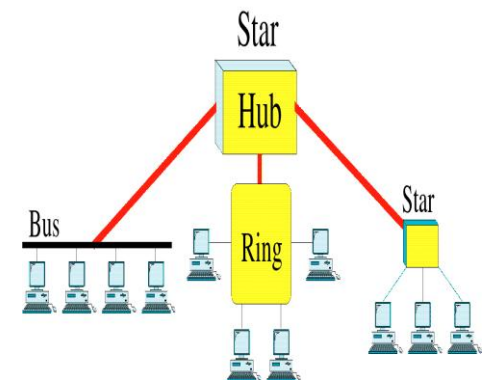
Bus



Ring



Hybrid



Network Software

PROTOCOL HIERARCHIES

Layers -

The concept that network software is organized functionally into levels. A level on one host talks to the same level on another host (its peer).

Protocol -

The protocol is the convention or standard that a layer uses to talk to the other layer. An agreement or standard on the conversation.

Physical Medium -

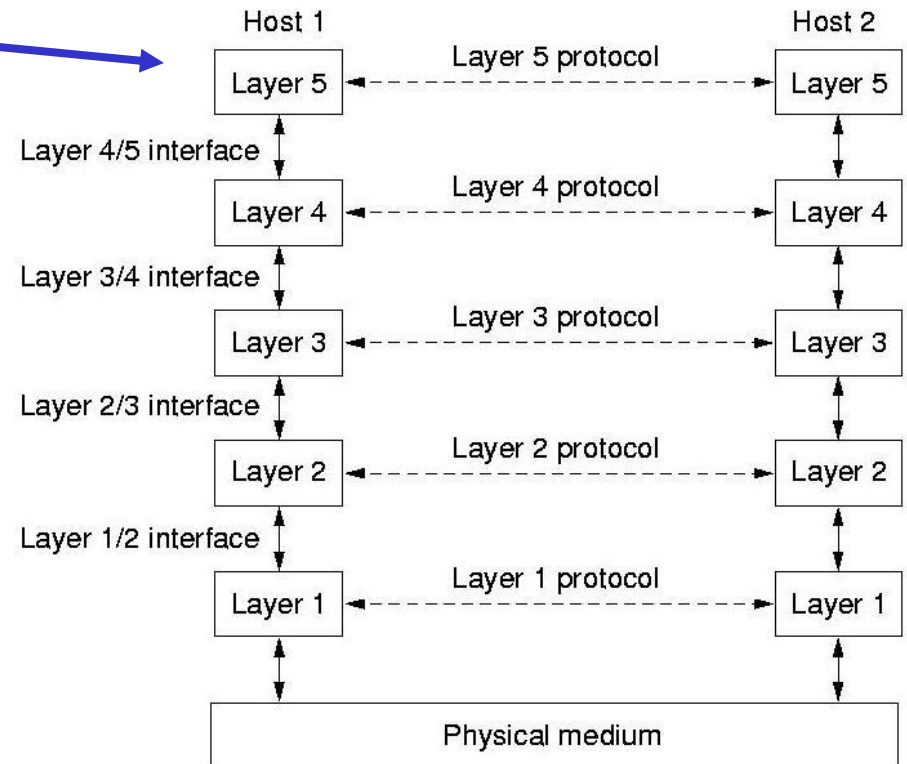
Underneath the layers is the wire or fiber or whatever.

Interface -

Defines the services that one layer offers another (either up or down.)

Important to keep this simple and clean.

Important that each layer perform specific actions.

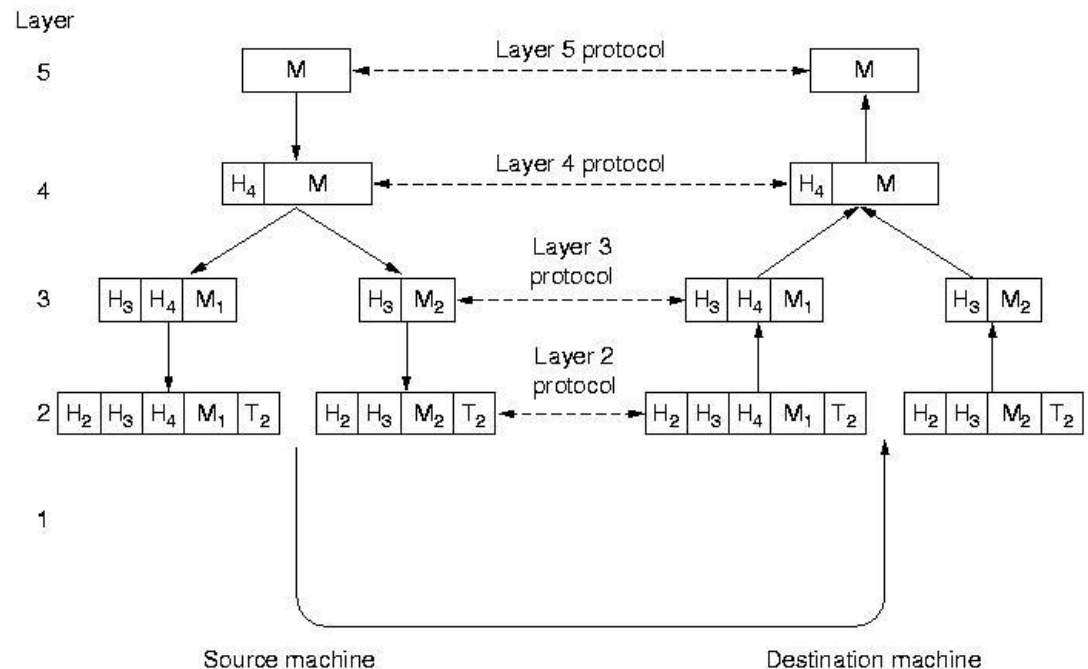


Network architecture -

- A set of layers and protocols. It contains details on what happens in the layer and what the layers says to its peer.

Protocol stack -

- A list of protocols used by a system, one protocol per layer.



INTERFACES AND SERVICES

Purpose of each layer is to provide services to the layer above it.

Entities / Peer entities -

Active element in each layer (process/IO chip).

Peer entity = layer N entity $\leftrightarrow$ layer N entity.

Layer N entity provides service for layer N + 1.

Service providers and users -

Layer N is a provider for user N + 1.

CONNECTION-ORIENTED / CONNECTIONLESS SERVICES:

Connection oriented service -

Like the phone system. The system establishes a connection, uses it, and closes it. Acts like a tube. Data comes out the other end in the same order as it goes in.

Connection Setup

Data Transfer

Connection Termination

Connectionless service -

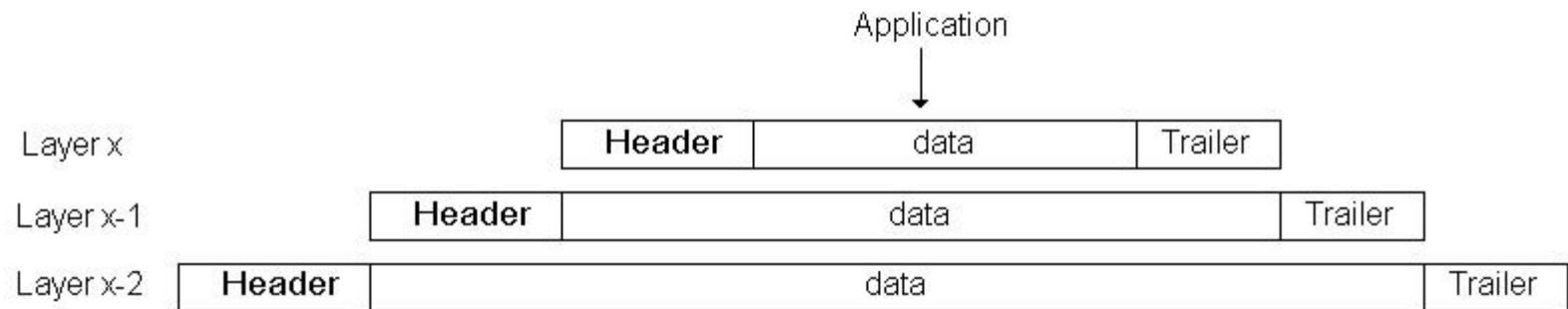
Like the post office. Each message has the entire address on it. Each message may follow a different route to its destination. Ordering not maintained.

Data Transfer

Headers, Data, and Trailers

flags	source	destination	priority	next protocol	data	CRC
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Encapsulation



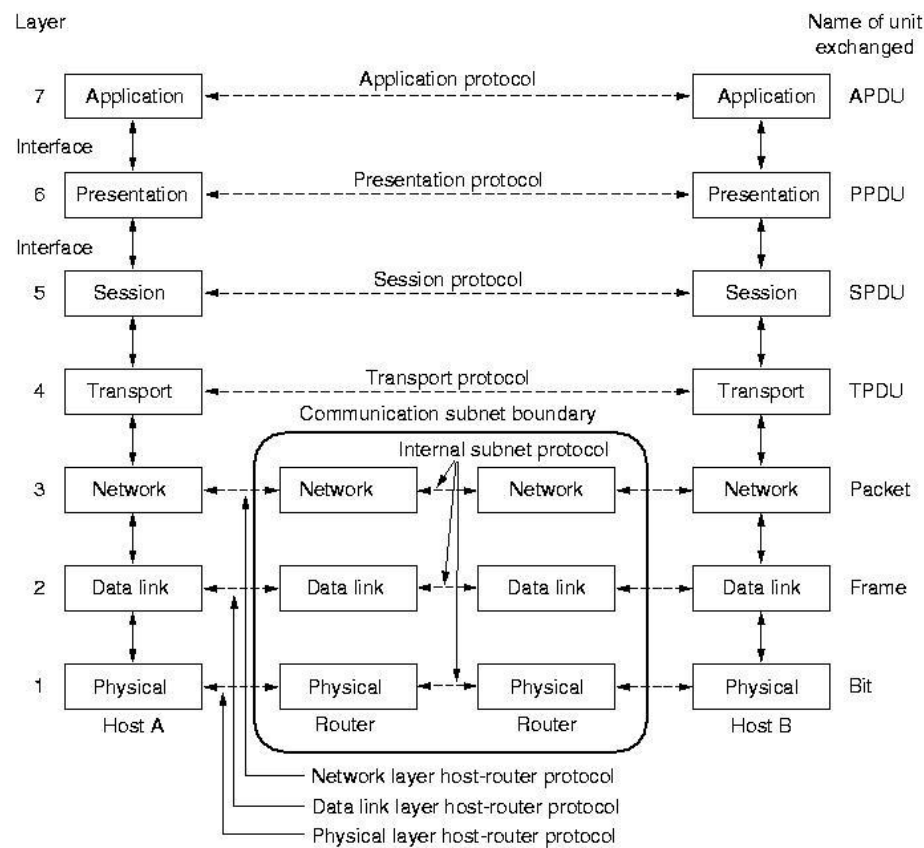
THE OSI REFERENCE MODEL

OSI == Open Systems Interconnection

Developed by **ISO** == International Standards Organization

Principles used to develop OSI Layering:

1. Need a layer for each different level of abstraction.
2. Each layer performs a well defined function.
3. Each layer should have standardized protocols.
4. The right number of layers - don't put too many functions together, but not too many layers either.



THE OSI REFERENCE MODEL

Physical Layer -

Purpose -- Transmits raw bits across a medium.

Representation of Bits – bits must be encoded into signals – electrical or optical

Electrical -- Concerns are voltage, timing, duplexing, connectors, etc.

Physical Topology -- star, ring, bus etc.

Transmission mode – simplex, half duplex, full duplex

Data Link Layer -

Framing -- Breaks apart messages into frames. Reassembles frames into messages.

Error handling -- solves damaged, lost, and duplicate frames.

Flow control -- keeps a fast transmitter from flooding a slow receiver.

Gaining Access -- if many hosts have usage of the medium, how is access arbitrated.

THE OSI REFERENCE MODEL

Network Layer -

- Routing** -- What path is followed by packets from source to destination. Can be based on a static table, when the connection is created, or when each packet is sent.
- Congestion** -- Controls the number packets in the subnet.
- Accounting** -- Counts packets/bytes for billing purposes.
- Heterogeneity** -- Interfacing so one type of network can talk to another.

THE OSI REFERENCE MODEL

Transport Layer -

Segmentation and reassembly – A message is divided into small pieces.

Reassemble the message correctly upon arriving at the destination.

Reliability -- Ensures that packets arrive at their destination. Reassembles out of order messages.

Service Decisions -- What type of service to provide; error-free point to point, datagram, etc.

Mapping -- Determines which messages belong to which connections.

Naming -- "Send to node xyzzy" must be translated into an internal address and route.

Flow control -- keeps a fast transmitter from flooding a slow receiver.

THE OSI REFERENCE MODEL

Session Layer -

Sessions -- Provides services that span a particular message. For instance, a login session could be logged.

Synchronization— Allows a process to add checkpoints into a stream of data. Provide way to subdivide a long mechanism for reliability.

Presentation Layer -

Prettiness -- Syntax and semantics of information transmitted. Understands the nature of the data being transmitted. Translation, Encryption, compression. Converts ASCII/EBCDIC, big endian/little endian

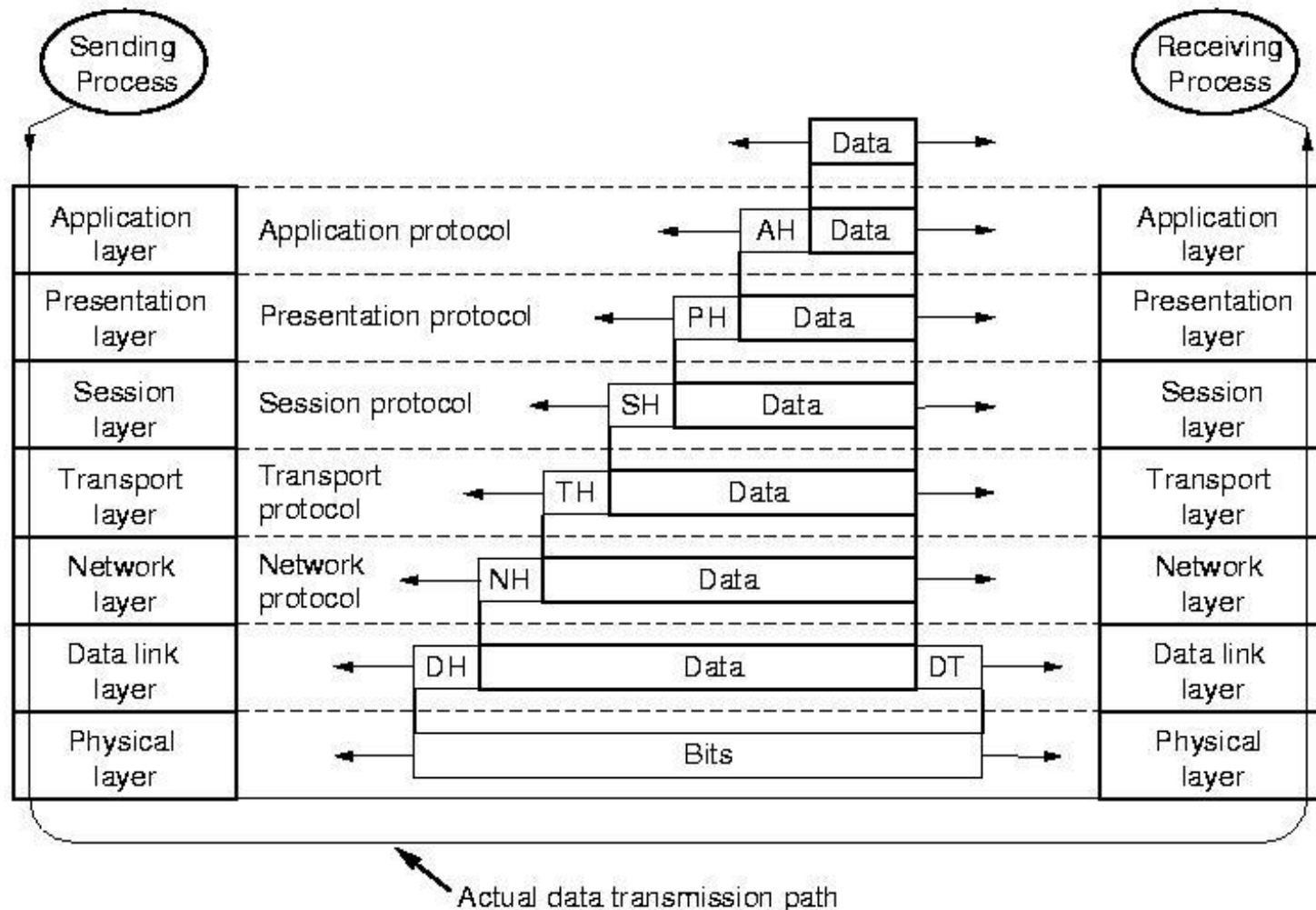
Application Layer -

Interfacing -- Terminal type translation.

File transfer -- Programs able to understand directory structures and naming conventions and map them onto various systems.

THE OSI REFERENCE MODEL

Data Transmission in the OSI Model -



THE TCP/IP REFERENCE MODEL

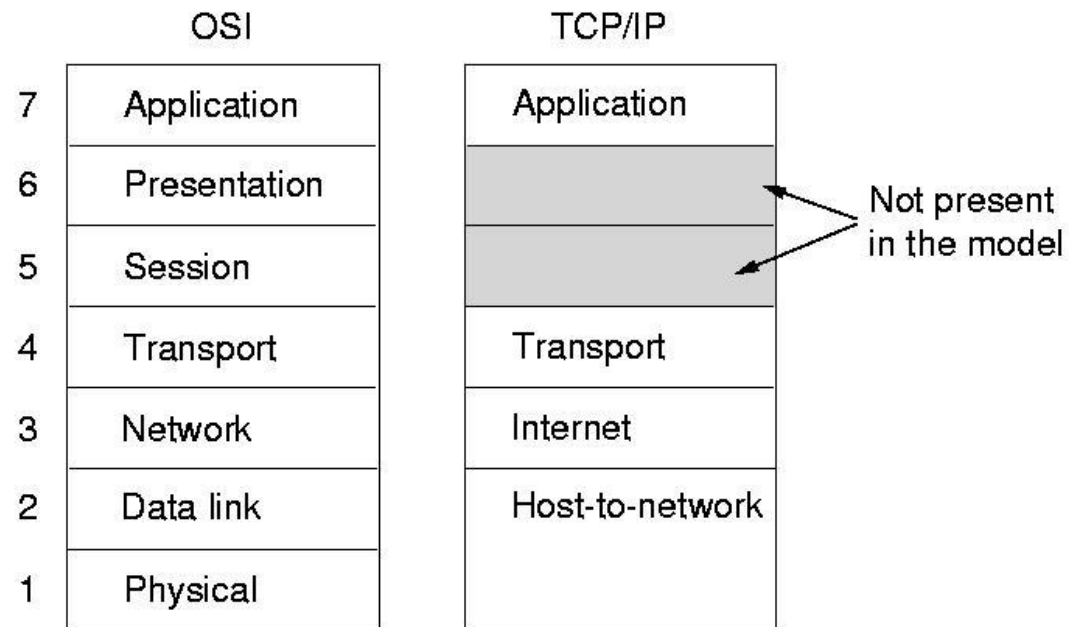
The TCP/IP protocol suite is made of four layers. Used in the Internet and ARPANET. Common mechanism that is gaining on/surpassing the OSI Model.

Host to Network Layer -

- This lowest level is not defined in this model. Various mechanisms abound.

Internet (Network) Layer -

- Provides packet switched connectionless service.
- Defines an official packet format and protocol called **IP (Internet Protocol)**.
- Deliver IP packets where they are supposed to go.
- congestion control.



Transport Layer -

Allows peer entities to communicate.

- TCP** -- **Transmission Control Protocol** provides a reliable connection oriented protocol that delivers a byte stream from one node to another. Guarantees delivery and provides flow control.
- UDP** -- **User Datagram Protocol** provides an unreliable connection-less protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own.

Application Layer –

It contains all the higher-level protocols.

- | | |
|-----------------|-----------|
| Terminal | -- Telnet |
| File transfer | -- FTP |
| The Web | -- HTTP |
| Electronic mail | -- SMTP |

COMPARISON BETWEEN OSI & TCP/IP REFERENCE MODELS

OSI has good definition of service, interface, and protocol as discussed before. Fits well with object oriented programming concepts. Protocols are better hidden.

With TCP, the protocols came first; model was just a description of the protocols. But then the model isn't good for any other protocols.

	LAYERS	
	Transport	Network
OSI	Connection-Oriented	Connectionless Connection-Oriented
TCP/IP	Connectionless Connection-Oriented	Connectionless

Example - Networks

NOVELL NETWARE:

Heavily used in PC world.

THE ARPANET:

1968 Originally intended as reliable network, with multiple routing.
Used TCP/IP precursor, which got built into early UNIX.

THE INTERNET:

Growing exponentially.

All nodes run TCP/IP. Means that all nodes have an IP address by which they can be contacted.

Services provided include:

- o e-mail
- o news
- o remote login
- o file transfer
- o the web

Example - Data Communications - X.25 NETWORKS

Public Networks

X.25 NETWORKS

First connection-oriented network developed during 1970's. Interface between public packet-switched networks and customers.

Operate at 64 Kpbs, so are very slow and becoming outdated. However there are still many of them in operation.

Connection oriented. Uses:

- **Switched Virtual Circuit** - established when the first packet is sent. Circuit remains for duration of session providing in-order delivery, and flow control.
- **Permanent Virtual Circuit** - established by agreement between the customer and the carrier. Like a leased line.

Example- Data Communications - FRAME RELAY:

FRAME RELAY:

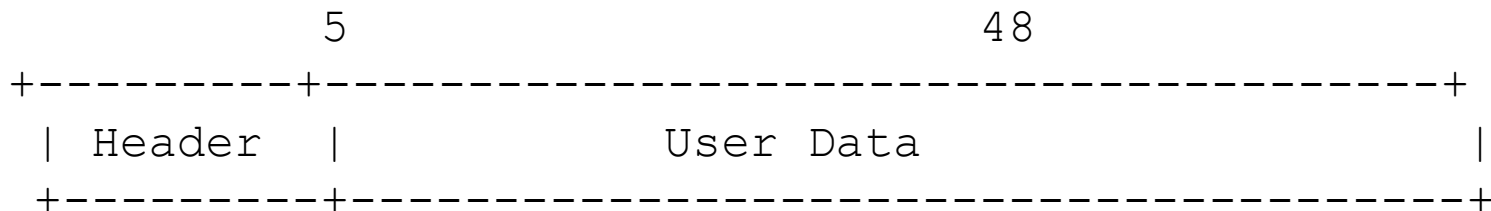
- It is connection-oriented network with no error control and no flow control.
- Runs at 1.5 Mbps.
- Uses permanent virtual circuit between two points.
- Takes advantage of modern high-speed reliable digital phone lines.
- This allows simple protocols with work done by user computers rather than by the network.

Example- Data Communications - ATM

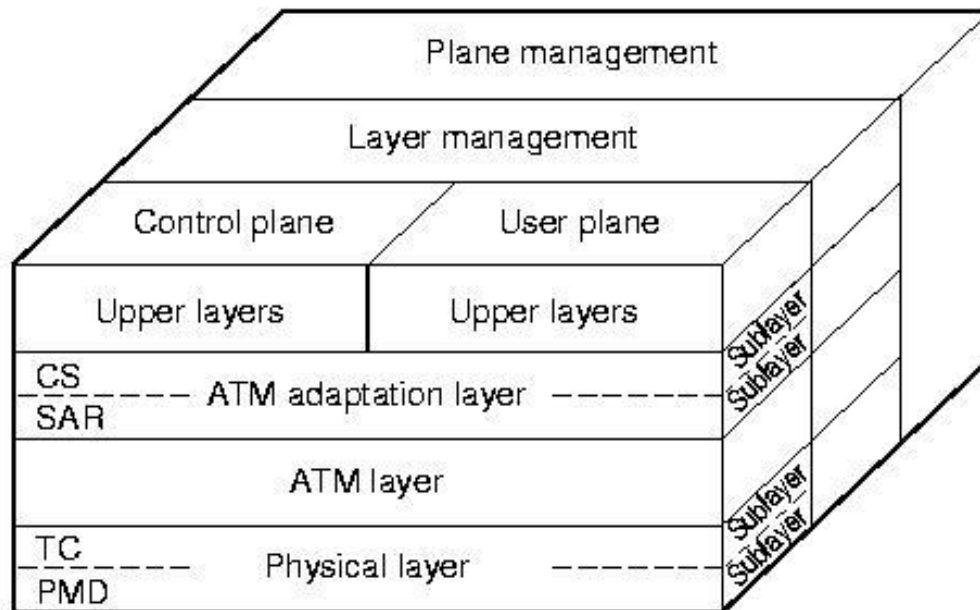
ISDN (Integrated Services Digital Network) will offer cable, video on demand, e-mail, etc.

ATM (Asynchronous Transfer Mode) –

- It is a connection-oriented packet switched network. Make connecting request first; then all cells follow the same path.
- Connections are called **virtual circuits**.
- Allows the transmission at 155 Mbps and 622 Mbps.
- Transmits in small fixed-size packets is called cells. Cells are 53 bytes long.



ATM Reference Model:



CS: Convergence sublayer
SAR: Segmentation and reassembly sublayer
TC: Transmission convergence sublayer
PMD: Physical medium dependent sublayer

The ATM layers and sub-layers, and their functions.

OSI layer	ATM layer	ATM sublayer	Functionality
3/4	AAL	CS	Providing the standard interface (convergence)
		SAR	Segmentation and reassembly
2/3	ATM		Flow control Cell header generation/extraction Virtual circuit/path management Cell multiplexing/demultiplexing
2	Physical	TC	Cell rate decoupling Header checksum generation and verification Cell generation Packing/unpacking cells from the enclosing envelope Frame generation
1		PMD	Bit timing Physical network access

Standards

Issues Include:

Phone companies and the governments that regulate them.

ISO (International Standards Organization)

ANSI (American National Standards Institute)

NIST (National Institute of Standards and Technology)

IEEE (Institute of Electrical and Electronics Engineering)

IAB (Internet Architecture Board)